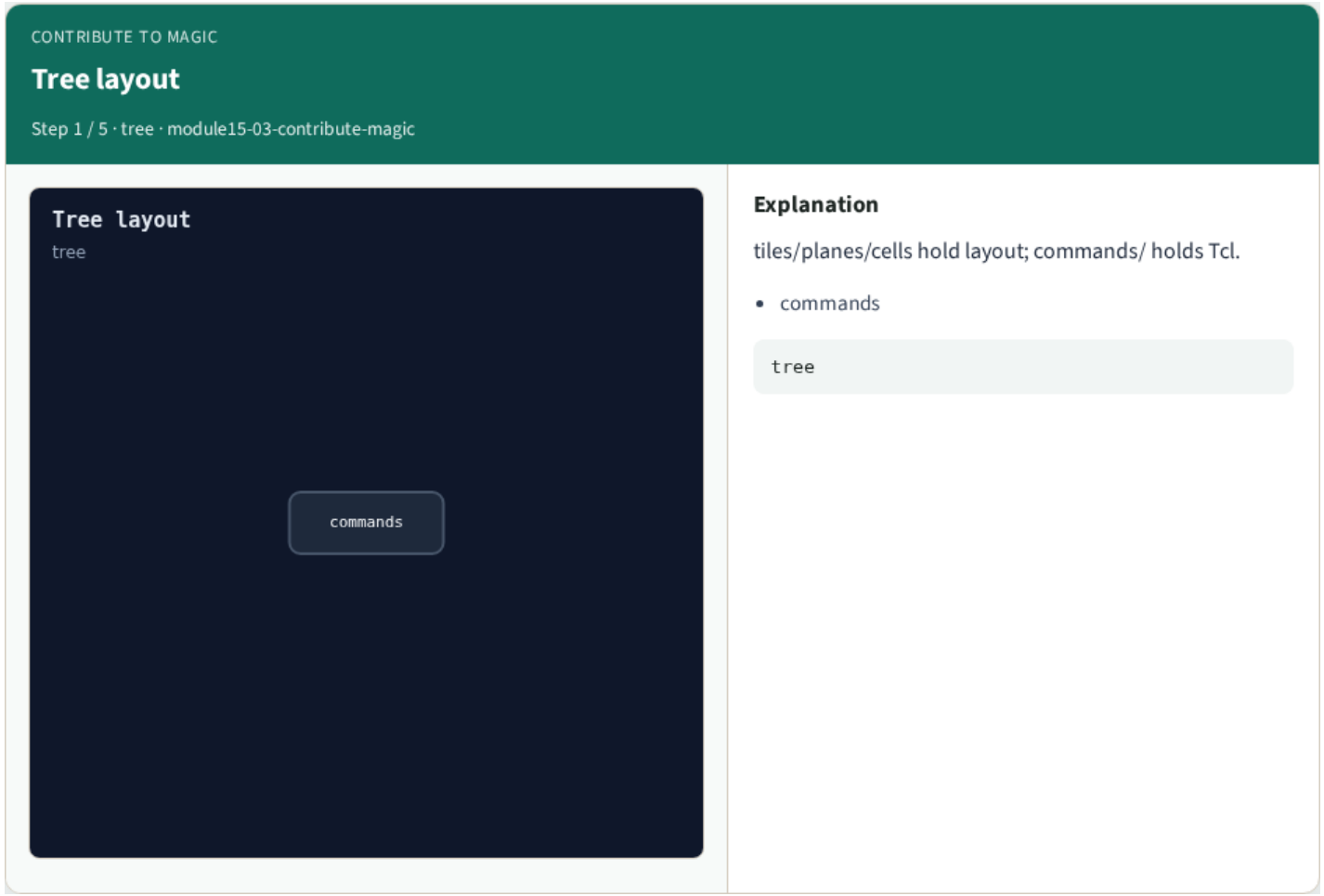
# Why Magic internals

Know where a new Tcl command lives and the minimal stub to build

# The idea

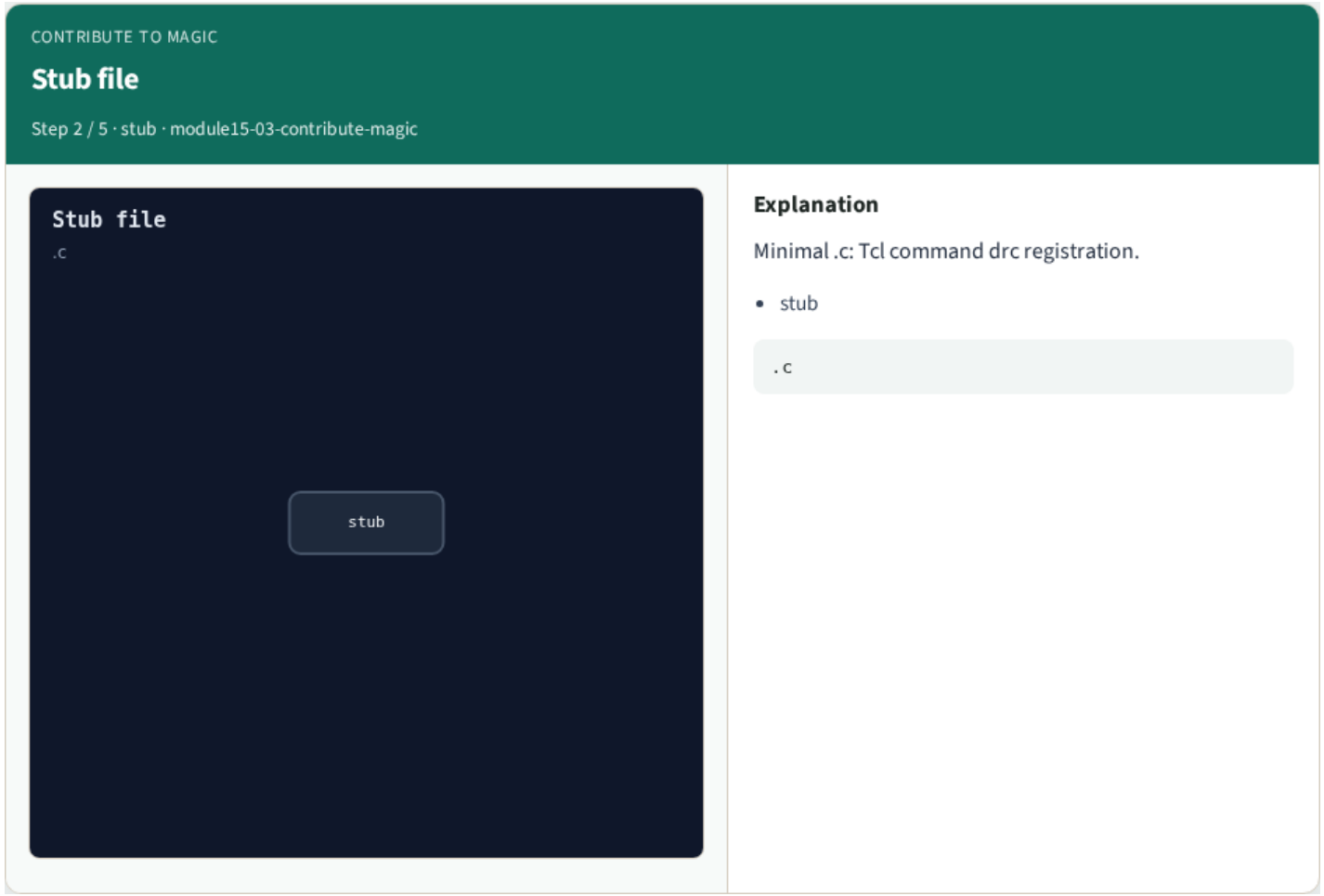
- GOLDEN: commands/cmd_drc.c + Tcl register + Makefile + magic -help drc

# Tree layout



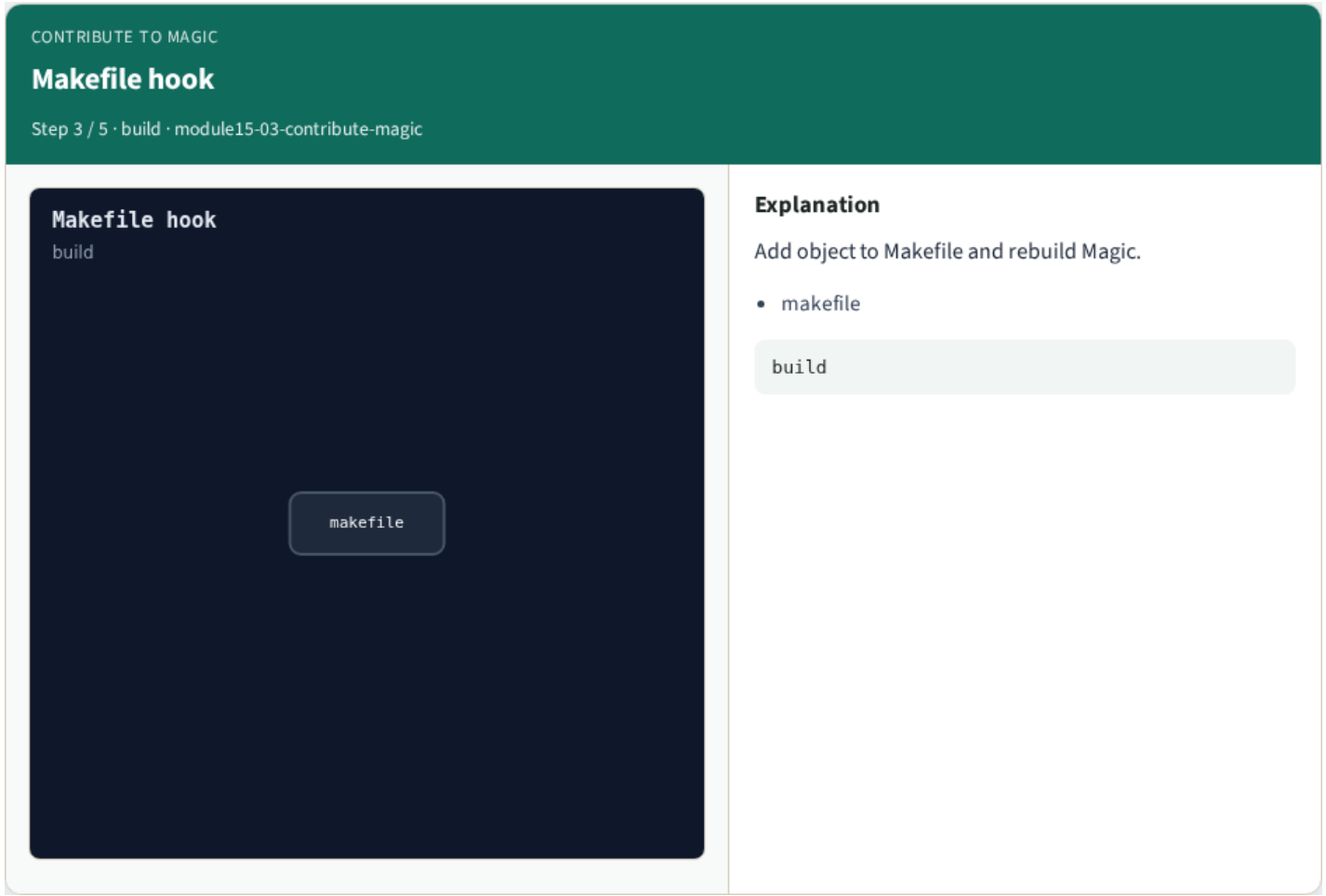

Tree layout

# Stub file



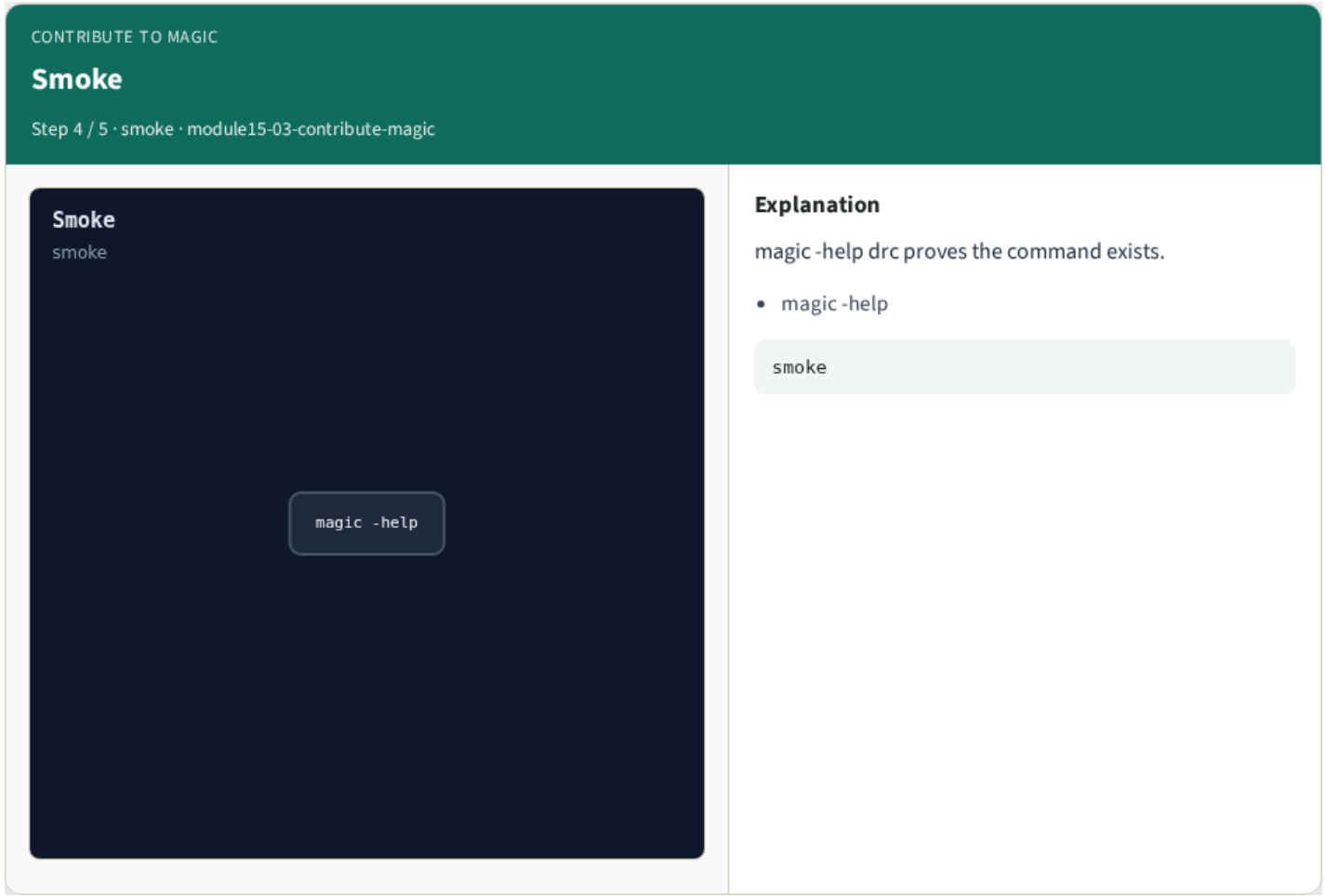

Stub file

# Makefile hook



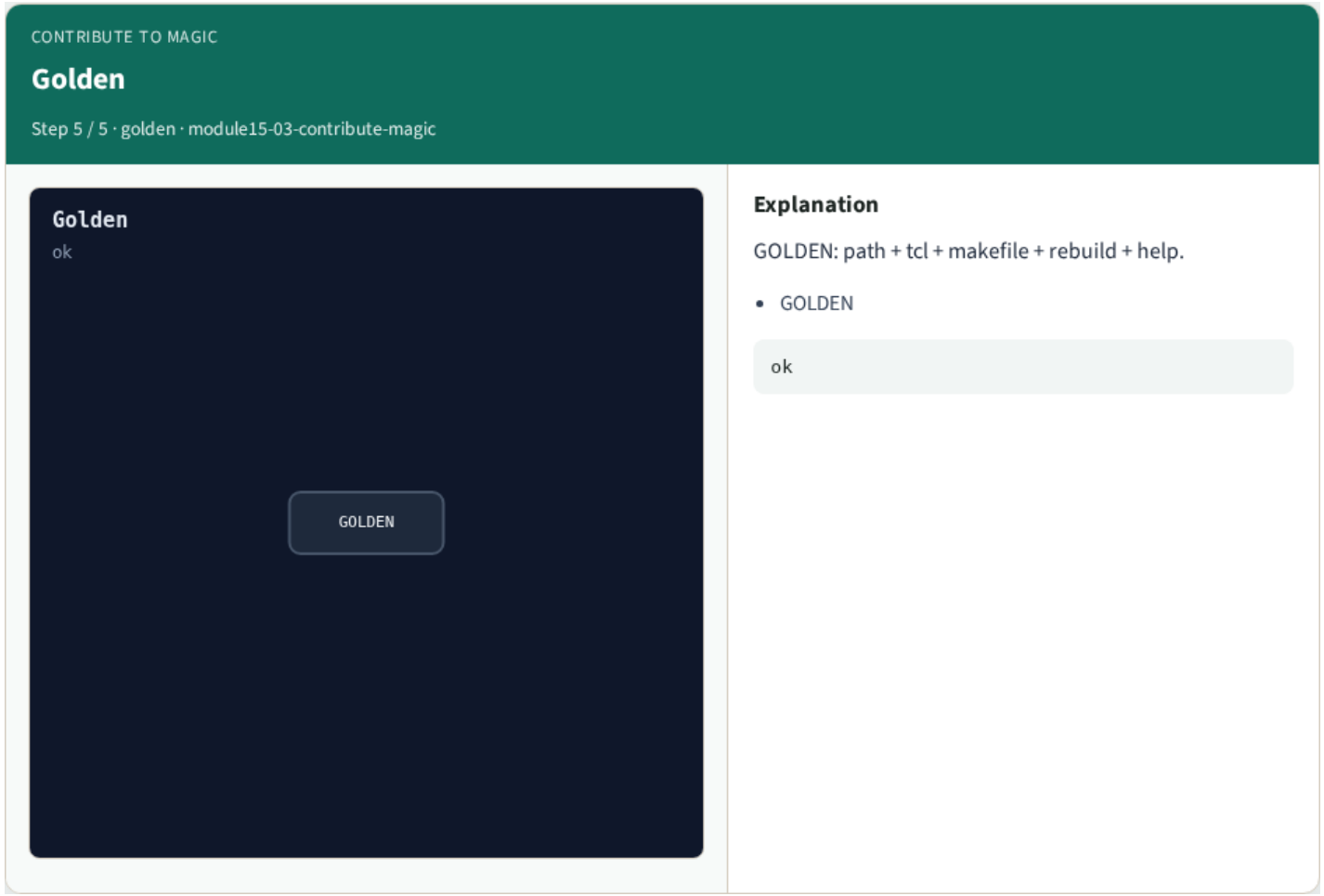

Makefile hook

# Smoke


CONTRIBUTE TO MAGIC
Smoke
Step 4 / 5 · smoke · module15-03-contribute-magic
Smoke
smoke
magic -help
Explanation
magic -help drc proves the command exists.
magic -help
smoke

Smoke

# Golden




Golden

# Browser lab track

- Open **contribute-magic**
- Click through stages and check goldens

# Implement track

- Call contribute_magic() in internals_solvers; assert golden match

# Pitfalls

- Treating browser JSON as the C API
- Confusing Tile with OpenSTA Graph

# Try these

- From the common folder, run the internals self-test

# Try these — try these

```
cd courses/learn_magic/common
python3 internals_solvers.py
```

# Pseudocode

- Here is the checklist in plain steps

# Pseudocode — try these

```
commands/cmd_drc.c + Tcl register + Makefile + magic -help drc.
```

# Next

- Complete the quiz for this part
- Wrap the Magic path , map, CLI evidence, algorithms, and C tree literacy